AI-Driven Multimodal Architectural Maturity Assessment for Modern SaaS Systems

Abstract

Modern SaaS systems rely on increasingly complex architectural patterns including security, multi-tenancy, reliability, observability, CI/CD automation, and cloud cost governance. While cloud providers and open-source communities have established comprehensive architectural frameworks, like the AWS Well-Architected Framework, the AWS SaaS Lens, Azure Cloud Adoption Framework (CAF), Google Cloud Architecture Framework, and the CNCF Cloud Native Maturity Model, the manual evaluation of a SaaS system against these standards remains labor-intensive, subjective, and heavily dependent on senior architectural expertise.

This project proposes a multimodal AI framework capable of automatically assessing the architectural maturity of SaaS systems by analyzing both textual descriptions and architectural diagrams. Using a combination of supervised learning, large language models (LLMs), computer vision–based diagram parsing, and rule-based reasoning, the system aims to assign maturity scores across key engineering dimensions and generate evidence-based recommendations for improvement. The work also lays the foundation for future extension into source code analysis, enabling a full-spectrum architectural evaluation pipeline.

1. Introduction

Modern SaaS architectures involve multi-dimensional decisions spanning application design, security, infrastructure, data isolation, deployment automation, observability, cost optimization, and operational excellence. As cloud-native ecosystems evolve, engineering organizations increasingly rely on structured architectural frameworks to guide system design and modernization.

2. Problem Statement

The central research question is:

> Can multimodal AI models interpret SaaS architecture documentation, both textual descriptions and visual architecture diagrams, and produce reliable maturity assessments aligned with established cloud and industry frameworks?

Sub-questions include:

- Can an AI system map architectural signals to maturity scores consistently?

- Can it extract meaningful semantics from architecture diagrams?
- Can it synthesize actionable recommendations grounded in recognized best practices?
- Can this framework later be extended to evaluate code-level maturity?

The overarching goal is to design and validate a prototype AI evaluator that assists engineering leaders in identifying architectural gaps and modernization opportunities.

3. Background and Motivation

The complexity of modern SaaS systems has increased dramatically due to the widespread adoption of:

- microservices and distributed communication
- multi-tenant data models
- Kubernetes and serverless compute platforms
- advanced CI/CD pipelines
- cloud-native observability stacks (metrics, logs, traces)
- DevSecOps and SRE practices
- FinOps cost optimization frameworks

As a result, organizations rely heavily on cloud architecture frameworks to ensure robustness, security, and efficiency. Yet adherence to these frameworks is uneven, largely because:

- expertise varies across teams
- documentation quality is inconsistent
- diagrams often contain implicit assumptions
- manual reviews do not scale
- architecture debt accumulates silently

A structured, AI-assisted evaluator can help mitigate these challenges by providing **consistent, repeatable, framework-aware insights**.

4. Data Requirements

The project leverages three primary sources of data.

4.1 Synthetic Multimodal SaaS Architecture Descriptions

A generator will produce diverse SaaS architectures by varying:

Textual Attributes

- tenancy and data isolation strategy
- authentication and authorization mechanism
- deployment topology (monolith, microservices, serverless)
- container orchestration models
- scaling and resilience mechanisms
- observability configuration
- compliance and audit features
- CI/CD practices
- cost governance and tagging strategies

4.2 Real-World Reference Architectures

Sample architecture documentation will be collected from authoritative sources, including:

- AWS Well-Architected and SaaS Lens reference diagrams
- Azure CAF architecture examples
- Google Cloud Architecture Center
- CNCF project architecture diagrams
- Open-source project documentation (e.g., GitLab, Hasura, Supabase)
- Technical whitepapers and conference presentations

These examples will be used primarily for evaluation and robustness testing.

4.3 Expert-Defined Maturity Rubrics

A structured rubric will map architectural characteristics to maturity scores.
A simplified excerpt is shown below:

Dimension	Score 0	Score 3	Score 5
Security	No auth; no TLS	Central IdP; basic RBAC	Strong IAM, isolation, encryption & audit
Multi-Tenancy	Shared DB	Schema isolation	DB-per-tenant; per-tenant metrics & throttling
Reliability	Single instance	Auto-scaling	HA, DR, chaos testing
Operations	Manual deployment	Basic CI/CD	GitOps; SRE practices; shift-left reliability
Cost Governance	No tagging	Basic cost centers	Automated insights & anomaly detection

These rubrics allow automated labeling of synthetic data and consistent evaluation of real architectures.

5. Methodological Approach

The proposed solution integrates **NLP, computer vision, and structured reasoning**.

5.1 Textual Analysis (NLP + Supervised Learning)

- Fine-tuning transformer-based models (e.g., BERT, RoBERTa, LLaMA variants)
- Classifying or regressing maturity scores
- Extracting architectural features using:
    - prompt-based LLM extraction
    - semantic feature identification
    - pattern recognition (e.g., “db-per-tenant”, “centralized IdP”, “Kubernetes”)

5.2 Diagram Understanding (CV + OCR + Graph Extraction)

The system will interpret architecture diagrams by:

- Recognizing cloud icons (database, load balancer, compute node, cache, etc.)
- Extracting text using OCR
- Identifying connections (arrows) and grouping components
- Transforming diagrams into machine-readable architecture graphs
- Cross-validating diagram-derived features with text-derived features

Potential techniques:

- CLIP or DINOv2 for icon/component detection
- Tesseract or transformer-based OCR
- Graph reconstruction algorithms

5.3 Hybrid Reasoning and Recommendation Generation

Using LLM reasoning, the system will:

- Justify each maturity score
- Detect contradictions between text and diagram
- Generate improvement recommendations aligned with:
    - AWS Well-Architected Pillars
    - SaaS Lens best practices
    - Azure CAF operational excellence guidelines
    - CNCF cloud-native maturity guidance

This creates a structured and actionable output.

5.4 Future Extensibility: Source Code Analysis

The system architecture will be designed to support future integration of:

- static code analysis
- AST-based quality evaluation
- CI/CD configuration evaluation (GitHub Actions, GitLab CI, ArgoCD)
- detection of anti-patterns, smells, vulnerabilities
- automated mapping from code insights → maturity scoring

This positions the project as a step toward a full **end-to-end architectural evaluation platform**.

6. Expected Contributions

This project aims to demonstrate:

Technical Contributions

- A multimodal AI pipeline integrating text and diagram analysis
- Reliable alignment with cloud-native maturity frameworks
- Cross-modal consistency checking
- Automated recommendation synthesis

Practical Contributions

- A functional tool to support SaaS architecture evaluations
- A reproducible rubric-based scoring methodology
- A foundation for real-world engineering decision support
- A path toward code-level maturity evaluation

Educational Contributions

- Demonstrated mastery of:
    - NLP

- Computer vision
- Diagram understanding
- Hybrid rule + ML systems
- AI reasoning applied to engineering domains

7. Conclusion

By integrating multiple AI modalities, textual understanding, diagram interpretation, and architectural reasoning, this project seeks to build a novel yet practical tool for assessing SaaS architectural maturity. The proposed system aligns directly with cloud provider best practices and cloud-native maturity models. If successful, it can serve as a prototype for real-world organizational tooling that helps engineering teams evaluate architectures more consistently and accelerate modernization efforts.